

1. What is a hypothesis? [1]

- A list of materials used in a lab.
- The final score of an experiment.
- A proven fact that never changes.
- **A testable prediction, often written as an 'If..., then...' sentence.**

• We measured 150 mL of water in a glass beaker.

• **How does water temperature affect the time it takes for sugar to dissolve?**

• Sugar dissolves faster in hot water than in cold water.

2. What is the independent variable in an experiment? [1]

- The variable that is measured at the end.
- **The one variable that the scientist changes on purpose.**
- The equipment used to measure data.
- A variable that stays the same to keep the test fair.

5. What is the dependent variable in an experiment? [1]

- A factor kept exactly the same in all trials.
- A rule in the lab to keep students safe.
- The variable that the scientist decides to change.

• **The variable that is measured or observed as the result.**

3. In the parachute experiment, what is the dependent variable? [1]

- **The time it takes for the parachute to fall to the floor.**
- The color of the plastic sheet.
- The size of the parachute canopy.
- The weight of the paperclip.

6. In a lab report, what do you do in the 'Analysis' section? [1]

• Guess what might happen before doing any tests.

• **Look closely at the data to find patterns and compare results.**

• Clean up the lab table and put away tools.

• Choose a new topic to study.

4. Which of the following is an investigation question? [1]

- The hypothesis was supported by our results.

7. Why do scientists repeat an experiment multiple times (doing several trials)? [1]

• To use up all the leftover lab materials.

- To change the independent variable each time.
- **To get reliable results and calculate an accurate average.**
- To make the experiment more confusing.
- They change randomly during every trial.
- **They make sure the experiment is a fair test.**
- They make the experiment finish faster.

8. In the ramp experiment, the ramp height was tested at 5 cm, 10 cm, 15 cm, and 20 cm. The ramp height is the: [1]

- Controlled variable.
- **Independent variable.**
- Control group.
- Dependent variable.

9. What is the main purpose of the 'Conclusion'? [1]

- To list the names of everyone in the group.
- **To state whether the results support the hypothesis and explain what was learned.**
- To measure the temperature of water.
- To write down the step-by-step procedure.

10. Why are controlled variables important in an experiment? [1]

- They are the main result we want to measure.